KisanSetu (MandiFlow)

Multi-Agent Planning, Coordination and Recovery System (MAPCRS) Dynamic APMC Agro-Logistics and Autonomous Recovery Engine · Advanced OOP Course Project · Idea Document (pre-implementation)

1. One-Paragraph Pitch

KisanSetu is a discrete-event, multi-agent simulation of Indian perishable-produce logistics. Farmers, transporters and mandis are autonomous agents with **private** state (reserve prices, cost structures, gate capacity). They plan, negotiate and commit to physical contracts through a mediator, and the system **recovers deterministically** when a highway stalls, a mandi price crashes, or a truck breaks down. Farmers interact through voice and one-click cards instead of dashboards. The project is deliberately not CRUD: every interesting behaviour emerges from agent interaction, time-dependent decay, and compensating rollback of irreversible commitments.

2. Why This Needs Multi-Agent Design (Defense Argument)

- **Information asymmetry:** each party holds data it will not reveal (farmer reserve price, trucker margin, mandi remaining capacity and auction book). A central optimizer cannot see them; negotiation is the only way to discover feasible deals.
- **Concurrency on scarce resources:** gate slots per hour, trucks and escrow deposits are contended. Two farmers can want the last slot at the same instant.
- **Irreversible physical and transactional locks:** a truck cannot be double-booked and a slot cannot be sold twice, so commit/rollback semantics are essential.
- **Time-coupled dynamics:** quality decays continuously, prices respond to arrivals, roads change. A one-shot LLM or static optimizer answers a snapshot, while the system must keep re-planning as the world moves.

3. Domain Model

Commodities: tomato, onion, chilli, each with its own decay rate λ, grade thresholds and base price curve. **Network:** a road graph of farm clusters, mandis (Azadpur, Ghazipur, Vashi, Kolar), processing plants (ketchup/food-processing) and highway junctions. Edge

weights (time, cost, toll, congestion) are mutable at runtime. **Time:** simulated clock. One tick is a fixed number of simulated minutes, and all state changes are driven by events on a priority queue.

4. Agents and Private State

Agent	Responsibility	Private state	Public interface
FarmerAgent / FPOAgent	Owns harvest batch; chooses destination	Reserve price, decay constant, deposit budget, risk tolerance	Call for proposals, accept/reject
TransporterAgent	Moves cargo on road graph	Cost per km, fuel, mechanical health, true ETA	Bid, status signals (EN_ROUTE, STALLED, FAILED)
APMCMandiAgent	Double-auction book, gate quota, price dynamics	Order book depth, buyer reserves, true yard congestion	Quoted price, slot offers
ProcessorAgent (plant)	Buys distressed produce for ketchup / processing	Intake capacity, price floor	Distress bids
LogisticsBroker	Mediator: runs CNP, escrow, contract commit/rollback	Contract ledger	Message routing only

Agents never hold references to each other. Every interaction is a typed message through the broker or event bus.

5. Autonomous Planning

Quality decay: $Q(t) = Q_0 \cdot \exp(-\lambda \cdot t \cdot \mu_temp)$, with $Q_0 = 1.0$.

Grade thresholds (tunable per commodity): Grade A ⩾ 0.85, Grade B ⩾ 0.65, Grade C ⩾ 0.40, Spoiled below. Price multiplier per grade (for example 1.0 / 0.75 / 0.45 / 0.0), so decay is felt through *grade steps*, which is what the State pattern models.

Net margin: NetMargin(m) = P_m(t_arrival) · Q(t_transit) · Volume − TransitCost(d_m) − MandiFees_m.

Refinements over the original draft

1. Use the *grade-based* price multiplier alongside (or instead of) raw Q, so the continuous decay maps onto commercial grades mandis really pay by.
2. P_m is a **forecast**, not a lookup. It is the current quote adjusted by the expected arrival-surge penalty (known inbound trucks to that mandi). This is what makes planning non-trivial.
3. Add an expected-risk term: NetMargin_risk(m) = NetMargin(m) − risk_aversion · Var(delay on route to m). It gives farmers a reason to prefer a closer, safer mandi over a distant high-price one.
4. μ_temp is a function of simulated ambient temperature (time of day, region, season), and optionally reduced for refrigerated trucks (a priced option in the transporter bid).
5. The farmer also evaluates the **processing plant** as a permanent candidate with a low but stable price, which forms the floor for the distress logic.

6. Decentralized Coordination

6.1 Tripartite Contract Net Protocol (managed by the broker)

1. **Announce:** the Farmer posts a task (crop, volume, ready time, deadline) to the broker.
2. **Call for proposals:** the broker fans out to transporters (freight bid with ETA) and to the shortlisted mandis (gate slot offer plus expected price).
3. **Bid:** each party replies with its private-state-derived offer and an expiry time.
4. **Award:** the Farmer ranks (transporter, mandi) bundles by net margin and awards one.
5. **Two-phase commit:** the broker **reserves** the truck, the gate slot and the farmer's deposit (prepare). If all three succeed, it commits and the contract is ACTIVE. Otherwise it rolls back whatever was reserved.

Bids have time-to-live, so a stale bid is rejected rather than honoured.

6.2 Mandi double auction and gate quota

- The mandi admits at most **N trucks/hour**. Slots are discrete resources reserved by contract.

- Buy-side and sell-side orders clear at a uniform price per auction round. The clearing price is shifted by a **supply-surge decay**: price falls as the ratio of arrivals to the mandi's absorption capacity rises (for example $P = P_{base} \cdot (1 - k \cdot \max(0, \text{arrivals/capacity} - 1))$).
- Arrival-time coordination: the broker staggers ETAs across slots so the yard never saturates.

7. Fault-Tolerant Recovery (Deterministic and Testable)

Every fault is injected at a **scripted simulated time** with a **fixed random seed**, so each scenario has one expected outcome. Triggers are numeric predicates, not vibes.

#	Fault	Detection predicate	Recovery strategy	Rollback performed	Expected end state
1	Highway stoppage (+7 h transit)	Predicted Q at arrival < commercial-viability threshold (Grade C floor)	**DistressAuctionRecovery**: plants within 30 km bid; highest net wins	Release mandi gate slot, refund unused freight escrow share	Truck diverted to plant; sale at processing price
2	Mandi glut (50 trucks, price −40%)	Quoted price < farmer reserve price **or** NetMargin of alternative > current by margin δ	**DynamicReroutingRecovery**: recompute edge weights, re-run planning over remaining candidates	Cancel old gate slot, re-issue reservation at the new mandi	Truck reroutes to best viable mandi
3	Engine failure	Truck state → FAILED (or heartbeat missed for k ticks)	**PeerHandoffRecovery**: closest idle truck with capacity, rendezvous point chosen to minimise added decay	Release failed truck lock, re-bind contract to the new truck, transfer manifest	Cargo continues on peer truck; ETA and decay recomputed

Recovery principles

- **Idempotent:** compensating actions can be re-run safely (guard with a contract version number).
- **Hysteresis:** reroute only if the improvement exceeds a threshold δ, so agents do not oscillate between mandis.

- **Bounded attempts:** at most a fixed number of reroutes per contract, after which the fallback is the nearest processor.
- **Every recovery is logged** as an event so tests can assert the exact sequence.

8. Object-Oriented Design Patterns (and Why Each Is Justified)

Pattern	Where	Justification to state in the defense
State	CropFreshnessState (GradeA → GradeB → GradeC → Spoiled), TruckState (IDLE, EN_ROUTE, STALLED, FAILED)	Behaviour (price multiplier, allowed transitions, permitted recovery) changes with state. Illegal transitions (Spoiled → GradeA) are made unrepresentable.
Strategy	IRoutePlanningStrategy (Dijkstra, SpoilageMinimizing, TollAverse), IPricingStrategy (fixed, surge-decay, double-auction)	Algorithms are swappable at runtime, so the same scenario can be replayed under different strategies for comparison.
Mediator	LogisticsBroker	Removes circular dependencies among Farmer, Transporter and Mandi. All coordination lives in one place.
Command + compensation	TransitContractCommand: execute(), compensate(), abort()	Multi-resource commitment with atomic rollback. Commands are logged so recovery can be replayed.
Adapter	VoiceKioskAdapter / IVoiceChannelAdapter	Translates vernacular strings, DTMF and button taps into typed domain commands. The engine never sees raw text.

Pattern	Where	Justification to state in the defense
Observer	TelemetryWatchdogListener, AudioSpeakerNotifier	Broadcasts stall alerts, price shocks and reroute decisions to the recovery logic, dashboard and voice feedback.

Patterns are not decoration. Each one has a **test that fails without it** (see Section 12).

9. Concurrency and Event-Driven Engine (Design-Level)

- **Single-threaded deterministic core:** a discrete-event scheduler with a priority queue ordered by (simulated time, sequence number). Agents process one message at a time from their own mailbox. This gives reproducible runs and removes most lock bugs.
- **Actor-style agents:** each agent owns its state and communicates only by messages. No shared mutable state between agents.
- **Resource arbitration at the owner:** the *mandi agent* is the sole authority on its slots and the *transporter agent* on its truck, so two farmers racing for the last slot are serialized by the mandi's mailbox. First message by (timestamp, sequence) wins, deterministically.
- **Optional parallel mode:** agents run on a thread pool with per-agent mailboxes (concurrent queues), but the broker still serializes contract commits. Used to demonstrate thread safety, with the deterministic mode as the reference.
- **Deadlock avoidance:** the broker acquires reservations in a fixed global order (truck, then slot, then deposit), uses timeouts on every reservation, and never holds one lock while waiting on another agent's reply.
- **Message envelope (conceptual):** messageId, correlationId (contract), sender, receiver, performative (CFP, PROPOSE, ACCEPT, REJECT, CANCEL, INFORM), payload, simulatedTimestamp, ttl.

10. Accessible Interface

Farmer-facing, three controls only

1. Green mic button: "Bol kar bechein" (speak to sell).

2. Produce status card: Green = fresh Grade A / sell, Amber = at risk / dispatch now, Red = spoiled.
3. Green tick: one-click contract confirmation.

VoiceKioskAdapter (simulated): takes input such as "50 crates tamatar bechna hai" (and Hindi/Hinglish variants), runs lightweight rule-based parsing (quantity, unit, crop, intent) with a confidence score, and emits a typed command such as CreateHarvestBatchCommand. Low confidence produces a one-tap confirmation card instead of silently acting. Units (crate, quintal, kg) are normalized to a single internal unit.

Audio feedback: reroute decisions are spoken in the farmer's language, e.g. "Aapka truck Azadpur ke bajay Ghazipur ja raha hai, bhaav behtar hai." In the simulation this is text-to-speech simulated as a console line plus a speaker icon.

Console dashboard (planned panels)

- Truck transit lanes with progress bars along the route.
- Crop freshness meters in green, amber and red.
- Live mandi price tickers with arrows and surge/glut flags.
- Gate-slot occupancy per mandi.
- Recovery event log, where injected faults and recoveries are shown in **red/bold**.

11. Edge Cases, Loopholes and Mitigations

Risk	Why it is a problem	Mitigation
Two farmers race for the last slot	Double booking	Slot owned by mandi agent; atomic reserve; loser receives REJECT and re-plans
Herd effect: all rerouted trucks pick the same alternate mandi	Creates a new glut	Broker applies expected-arrival penalty in the forecast and staggers reroutes; hysteresis threshold
Ping-pong rerouting	Truck oscillates between mandis, decaying	Reroute only if gain > δ; cap on reroute count
Partial commit (truck reserved, slot fails)	Leaked locks	Two-phase commit with compensating release; reservation TTL

Risk	Why it is a problem	Mitigation
Compensation itself fails	Stuck resource	Idempotent compensate with retry and a dead-letter queue; invariant checker
Breakdown with no idle truck in range	Peer handoff impossible	Fallback chain: nearest truck finishing a trip, then local cold storage, then distress auction
Handoff rendezvous adds too much decay	Handoff worse than distress sale	Compare handoff net margin with distress net margin and pick the better
Strategic misreporting (farmer inflates reserve, mandi hides capacity)	Breaks auction fairness	Use sealed-bid second-price (Vickrey-style) rules where appropriate, and note the assumption honestly
Stale telemetry	Wrong recovery decision	Timestamp and TTL on every reading; heartbeat missed for k ticks counts as FAILED
Unrealistic assumption: perfect price forecast	Planning looks too good	Add noise to forecasts and report forecast error in evaluation
Unrealistic assumption: instant, lossless messaging	Hides distributed-systems issues	Optional message delay/drop in the event bus, with at-least-once delivery and idempotent handlers
Voice misrecognition	Wrong quantity or crop sold	Confidence threshold plus confirmation card; never commit from voice alone
Determinism vs. thread safety trade-off	Examiners may ask which is the "real" mode	Deterministic mode is the reference; parallel mode must reproduce the same final ledger within defined tolerance

12. Evaluation and Defense Plan

Measured outcomes (compare against a static baseline: one fixed mandi, no recovery)

- Average net margin per farmer, and total spoilage loss.
- Percentage of cargo salvaged after each injected fault.
- Yard congestion (peak arrivals vs. gate quota).
- Number of reroutes and oscillations.

Deterministic scenario tests

1. Fault 1 produces the diversion to a plant within 30 km and releases the gate slot.
2. Fault 2 produces a reroute to the expected alternate mandi.
3. Fault 3 produces handoff to the expected peer truck with a correct manifest.
4. Rollback leaves no leaked locks (invariant: reserved resources = resources owned by active contracts).
5. Racing slot requests give the same winner on every run.

Pattern-necessity checks: swapping strategies changes results without touching agents; removing the broker reintroduces circular dependencies; illegal state transitions are rejected.

13. Scope Tiers (to Protect the Schedule)

- **Must have:** three agents plus broker, decay and net margin planning, CNP with 2PC, three fault scenarios, all six patterns, deterministic scheduler, CLI dashboard.
- **Should have:** processor agent, voice adapter with a small vernacular lexicon, forecast surge penalty, hysteresis.
- **Could have:** parallel mode, message loss simulation, refrigerated truck option, multiple crops at once.

14. Proposed Structure and Phases

```
agro_multiagent/
├── core/        (Agent, Message, EventBus, Scheduler)
├── domain/      (Commodity, CropState, RoadGraph)
├── agents/      (FarmerAgent, APMCMandiAgent, TransporterAgent,
ProcessorAgent, LogisticsBroker)
├── commands/    (TransitContractCommand, CreateHarvestBatchCommand)
├── ui_adapter/  (IOrderInputPort, VoiceKioskAdapter, AudioSpeakerNotifier)
├── recovery/    (IFailureRecovery, DistressAuctionRecovery,
DynamicReroutingRecovery, PeerHandoffRecovery)
└── simulation/  (MainSimulation, FaultInjector, Dashboard)
```

Phases: (1) domain models and decay, (2) scheduler, messaging and CNP bidding, (3) command/rollback and mandi auction, (4) recovery strategies and fault injector, (5) voice adapter and dashboard, (6) evaluation and defense polish.

15. Open Decisions Before Implementation

- **Language:** Decided: C++20. RAII guards make lock and reservation rollback exception-safe, which fits TransitContractCommand compensation, and performance headroom helps the parallel mode.
- Number of mandis, trucks and farmers in the demo map (suggested: 4 mandis, 2 plants, 6 trucks, 5 farmers).
- Whether the double auction uses sealed-bid or continuous matching.

Next step when you are ready: the C++20 class and interface specification can be derived directly from Sections 4–9.